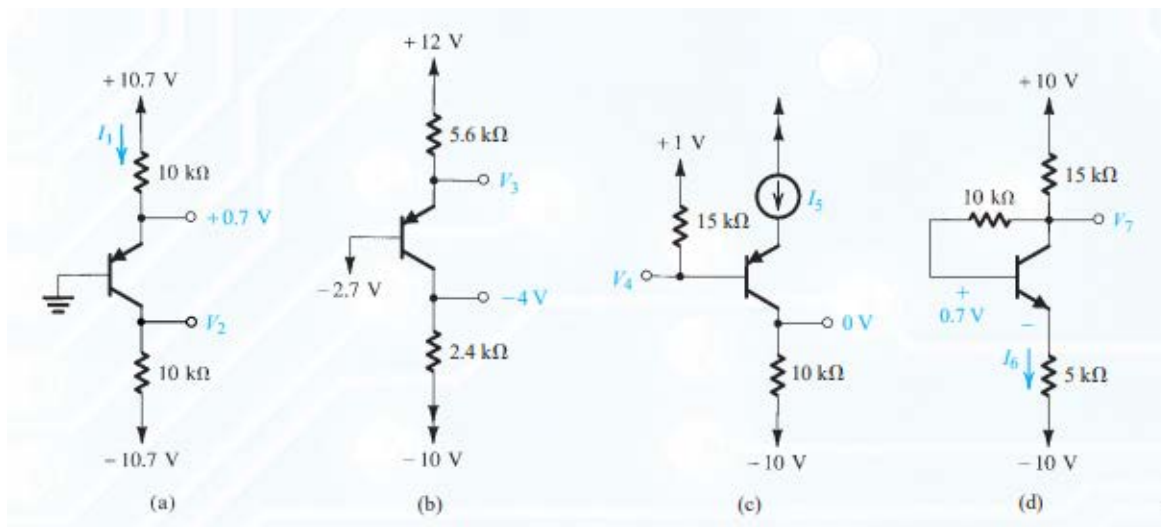


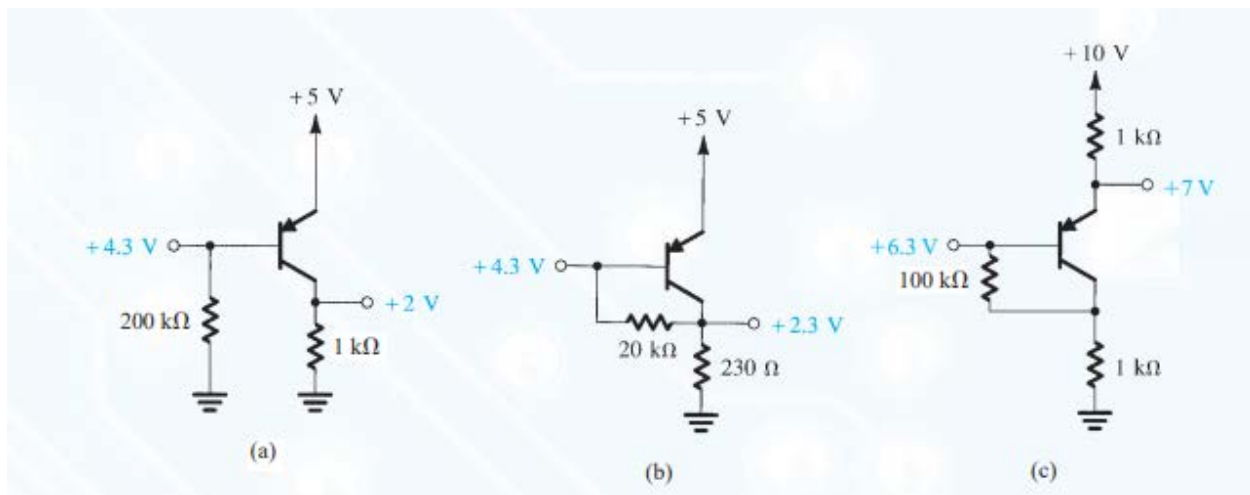
EE1004 – Electronic Devices & Circuits

Assignment#2 (CLO 2)

Question#1: For the circuits in Fig, assume that the transistors have $\beta=100$. Some measurements have been made on these circuits, with the results indicated in the figure. Find the values of the other labeled voltages and currents.



Question#2: Measurements on the circuits of Fig produce labeled voltages as indicated. Find the value of β for each transistor.

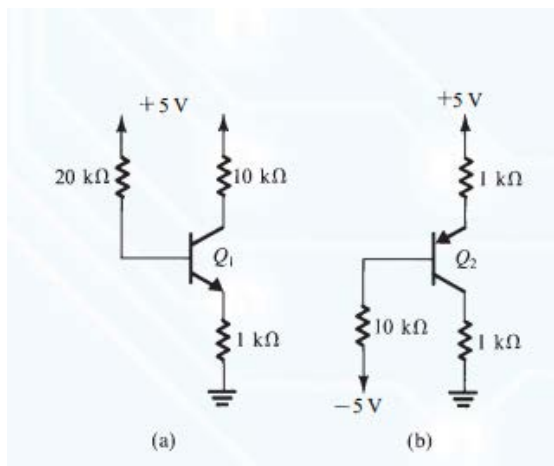


Question#3:

- Sketch the $i_c - v_{CE}$ characteristic curves of an npn transistor for $i_B = 1\mu A, 10\mu A, 20\mu A, 30\mu A$ and $40\mu A$. Assume $\beta = 100$.
- Sketch and label the load line on the graphs of part (a) for $V_{CC} = 5V$ and $R_C = 1k\Omega$.
- What peak to peak collector voltage swing will result for i_B varying over the range of $10\mu A$ to $40\mu A$.
- If $V_{CE} = \frac{1}{2}V_{CC}$, find the value of I_C and I_B .
- If at this current $V_{BE} = 0.7V$ and $R_B = 100k\Omega$, Find the required value of V_{BB} .

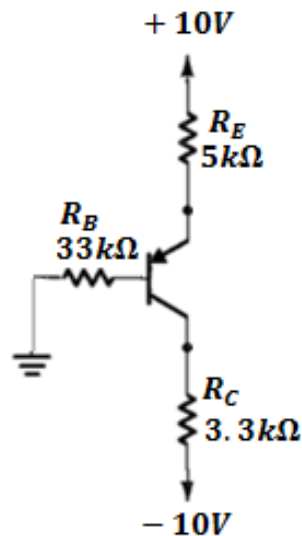
Question#4:

Find approximate values for the collector voltages in the circuits of Fig. Also, calculate forced β for each of the transistors. (Hint: Initially, assume all transistors are operating in saturation, and verify the assumption.



Question#5:

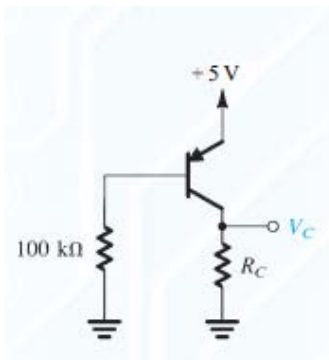
In the circuit shown, the transistor has $\beta = 50$.



- Find the values of V_B , V_E and V_C ?
- Find the operating region of BJT?
- Will there be any effect on operating mode if we change R_C from $3.3\text{ k}\Omega$ to $33\text{ k}\Omega$?
Explain with reason.
- Find the maximum or minimum (whatever is applicable) value of R_C that keeps the transistor in the active mode?

Question#6:

The pnp transistor in the circuit has $\beta = 50$. Find the value for R_C to obtain $V_C = +3\text{ V}$. What happens to circuit parameters and operating mode if the transistor is replaced with another having $\beta = 100$?



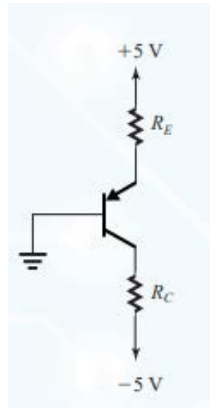
Question#7:

For the common emitter circuit with $V_{CC} = +10V$ and $R_C = 1K\Omega$, find V_{CE} and the voltage gain at the following dc collector bias currents: 1mA, 5mA, 8mA and 9mA. For each, give the maximum possible positive and negative output signal swing as determined by the need to keep the transistor in the active region. Show all your calculations and present your results in the following table.

Sr#	Ic(mA)	V_{CE}	Voltage gain A_V	Positive signal swing	Negative signal swing
1	1				
2	5				
3	8				
4	9				

Question#8:

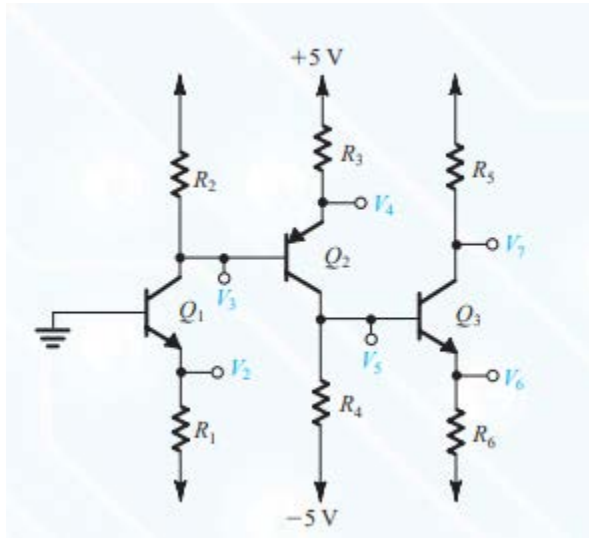
Design the circuit in Fig to establish a current of 1 mA in the emitter and a voltage of -1 V at the collector. The transistor $V_{BE} = 0.64V$ at $I_E = 0.1 mA$, and $\beta = 100$. To what value can R_C be increased while the collector current remains unchanged?



Question#9:

Using $\beta = \infty$, design the circuit shown in Fig so that the bias currents in Q_1 , Q_2 , and Q_3 are 1 mA, 1 mA, and 2 mA, respectively, and $V_3 = 0$, $V_5 = -2$ V, and $V_7 = 1$ V.

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Question#10:

The pnp transistor in the circuit in Fig has $\beta = 50$. Show that the BJT is operating in the saturation mode and find β_{forced} and V_C . To what value R_B should be increased in order for the transistor to operate at the edge of saturation?

